

Computer Vision:

Week-13 (27 April--3 May) Theory Lectures

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- **Semantic and Instance Segmentation.**

Contents

- Semantic vs Instance Segmentation
- U-Net and FCN Concepts
- Mask R-CNN
- Idea Losses: Cross-Entropy, Dice, and IoU Loss

Image Segmentation

- Image segmentation means dividing an image into meaningful parts or regions.
- **Why do we need it?**
- Simple classification tells us what is in an image
- Segmentation tells us where exactly it is — at the pixel level

Example

In a photo of a street, segmentation can identify every pixel as: road, car, person, building, sky

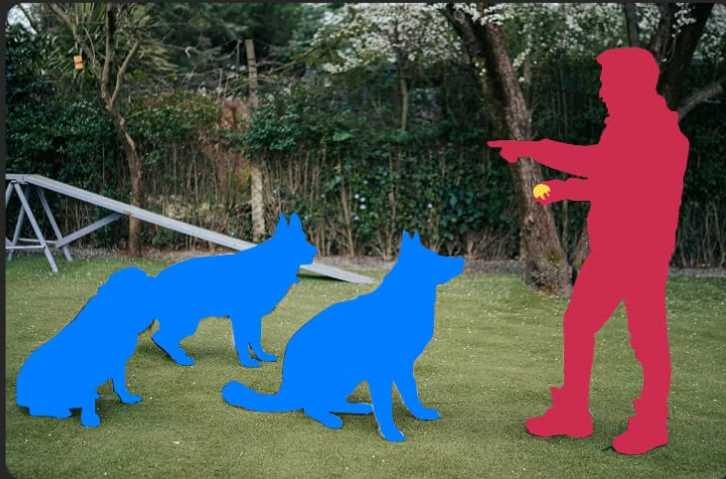
Two Types of Segmentation

Instance Segmentation

Semantic Segmentation

- Labels every pixel with a class
 - Does NOT separate individual objects
- Labels every pixel AND separates each individual object
 - Each object gets its own unique label

COMPARING SEMANTIC & INSTANCE SEGMENTATION APPROACHES



SEMANTIC SEGMENTATION

- Assigns a class label to each pixel
- Does not differentiate b/w objects of the same class
- Example: All cars in an image are labeled simply as "car"
- Produces one mask per class

USE CASES

- Road layout (lanes, sidewalks)
- Organ/tissue classification
- Land cover mapping
- Crop vs. weed detection
- Surface defect detection
- Scene parsing (AR/VR)



INSTANCE SEGMENTATION

- Assigns class label & unique instance ID to each pixel
- Differentiates b/w multiple objects of the same class
- Example: Each car gets its own label
- Produces one mask per instance

USE CASES

- Cars, pedestrians, cyclists
- Cell/tumor separation
- Individual houses/vehicles
- Fruit/plant counting
- Multiple defect labeling
- Person tracking

Semantic Seg:

What it does:

- Every pixel in the image is assigned a class label
- Example classes: person, road, tree, sky, car

Key Point:

- If there are 3 people in the image, all 3 are labeled as "person"
- It does NOT tell them apart individually
- Use Cases:
 - self-driving cars (understanding road layout)
 - Medical imaging (finding tumor regions)
 - Satellite image analysis

Output:

- A colored map where same-class pixels have the same color

Instance Segmentation

- Like semantic segmentation
BUT also separates each
individual object
- Each object instance gets a
unique label

Key Point:

- If there are 3 people → Person 1,
Person 2, Person 3 are all
separate
- It answers: what class? AND
which individual object?

Semantic Vs Instance

Feature	Semantic Segmentation	Instance Segmentation
Pixel labeling	Yes	Yes
Separates objects	No	Yes
Same class objects	Same color	Different colors
Complexity	Lower	Higher
Example model	FCN, U-Net	Mask R-CNN

Segmentation models

- **FCN-Fully Convolutional Network**

- Labels every single pixel in an image with a class name.
- All pixels of the same class get the same color — it cannot tell two people apart.

U-Net-Encoder-Decoder with Skip Connections:

- Same as FCN — labels every pixel — but produces much sharper and more accurate segmentation, especially on small or detailed objects like cells or tumors

Segmentation models

- **Mask R-CNN-Region-based CNN with Mask Branch**
- Detects each object individually, draws a bounding box around it, and also creates a pixel-level mask for it.
- Each object gets its own unique mask..

Segmentation models

Model	Think of it as...	Best for
FCN	Painting every pixel a color	General semantic segmentation
U-Net	FCN + memory shortcuts (skip connections)	Medical imaging, small datasets
Mask R-CNN	YOLO + pixel mask for each object	Instance segmentation, counting objects

N1:

- *An FCN model takes an input image of size **512 × 512**. The encoder applies 4 max-pooling layers (each reduces size by half). The decoder then upsamples back to original size. Calculate:*
- The size of the feature map after each max-pooling layer
- The final output size after upsampling?

N2:

- *A Mask R-CNN model processes an image and the RPN proposes 8 candidate regions. After filtering, only regions with confidence score above 0.5 are kept. The confidence scores are: 0.92, 0.45, 0.78, 0.61, 0.33, 0.85, 0.50, 0.67*
- How many regions are kept after filtering?
- What is the average confidence score of the kept regions?
- If each kept region produces a 28×28 pixel mask, what is the total number of mask pixels generated?